SIMATS ENGINEERING

ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

Name	**1.M.Kyathi Sai Priya** **2.S.Blessia** **3.Ashwini E B**
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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) Question Count: 5	**Total Marks: 50**

Code of Conduct

We 1. M.Kyathi Sai Priya (192572101) 2. S.Blessika(192525251) 3. Ashwini(192511422)___certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _M.kyathi

S.Blessika

Ashwini E B

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example

1. How virtualization enabled the growth of cloud service providers

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

How Virtualization Enabled the Growth of Cloud Service Providers

CSA1522-Cloud Computing & Big Data Analytics

CO1_AT2_TECHNICAL PRESENTATION

GUIDED BY

P.RAJARAM

PRESENTED BY

Blessika s(1925252

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KYATHI SAI PRIYA MEKAL

Introduction

- Virtualization is the technology that allows multiple virtual machines to run on a single physical server. It has become the foundation of cloud computing by improving resource utilization and reducing infrastructure costs.
- Introduces virtualization technology.
- Enables efficient use of hardware.
- Supports multiple virtual machines.
- Forms the basis of cloud computing.
- Helps cloud providers grow rapidly.

Objectives

- Understand the concept of virtualization.
- Explain how virtualization supports cloud service providers.
- Study the role of hypervisors and virtual machines.
- Identify the benefits of virtualization in cloud computing.
- Learn real-world applications of virtualization.
- Analyze how virtualization improves scalability and resource utilization.
- Understand how virtualization reduces infrastructure and operational costs

Problem Statement

- Traditional infrastructure requires separate physical servers for each application.
- Low hardware utilization increases operational costs.
- Scaling resources is slow and difficult.
- Managing multiple physical servers is complex.
- Cloud providers need a cost-effective, scalable, and efficient solution.
- Virtualization solves these challenges by enabling multiple virtual machines on a single server.

Applications

- **Cloud Computing** – Provides virtual servers and storage services.
- **Data Centers** – Improves hardware utilization and resource management.
- **Web Hosting** – Hosts multiple websites on a single physical server.
- **Software Testing** – Runs different operating systems for testing applications.
- **Disaster Recovery** – Enables quick backup and recovery of virtual machines.

What is Virtualization?

Definition

- Virtualization is the creation of virtual versions of physical resources.
- A hypervisor creates and manages multiple Virtual Machines (VMs).
- Each VM runs its own operating system independently.
- Multiple VMs share one physical server efficiently.

Advantages

- Better resource utilization
- Lower hardware cost
- Easy management
- Improved flexibility

How Virtualization Enables Cloud Computing

Working Process

- Physical servers host a hypervisor.
- The hypervisor creates multiple Virtual Machines.
- Each VM runs applications independently.
- Users access cloud services through the internet.
- Resources can be allocated or removed on demand.

Benefits

- **Scalability:** Resources can be increased or decreased quickly based on user demand.
- **High Availability:** Services continue to run even if one server fails.
- **Cost Efficiency:** Multiple virtual machines share one physical server, reducing costs.
- **Faster Service Deployment:** New virtual machines can be created and deployed in just a few minutes

Architecture Diagram

Users

|

▼

Cloud Platform

|

▼

Hypervisor (VM Manager)

▼	▼	▼
VM 1	VM 2	VM 3
Linux	Windows	Linux

▼

Physical Server

CPU • Memory • Storage

Real-World Use Case

Amazon Web Services (AWS EC2)

- AWS uses virtualization to provide virtual servers (EC2).
- Customers launch VMs within minutes.
- Resources can be scaled up or down based on demand.
- Multiple users securely share the same physical hardware.
- Businesses pay only for the resources they use.

Advantages of Virtualization

- **Efficient Resource Utilization:** Multiple virtual machines share one physical server, improving hardware usage.
- **Cost Reduction:** Reduces hardware, electricity, and maintenance costs by using fewer physical servers.
- **Scalability:** Resources can be increased or decreased quickly based on user demand.
- **Fast Deployment:** Virtual machines can be created and deployed within a few minutes.
- **Improved Reliability:** Virtual machines can be easily backed up and restored during failures.

Disadvantages of Virtualization

- **High Initial Cost:** Setting up virtualization infrastructure requires significant investment.
- **Performance Issues:** Running too many virtual machines may reduce system performance.
- **Complex Management:** Virtual environments require skilled administrators for maintenance.
- **Security Risks:** Misconfigured virtual machines or hypervisors can increase security vulnerabilities.
- **Single Server Failure:** Failure of the host server can affect all virtual machines running on it.

Conclusion

Conclusion

- Virtualization is the foundation of cloud computing.
- It enables efficient hardware sharing.
- It reduces infrastructure costs.
- It provides scalability, flexibility, and reliability.
- Cloud providers use virtualization to deliver services worldwide.

Thank you


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